5.1 Balancing chemical equations

Science understanding

Logical/Mathematical

Atoms are not created or destroyed in a chemical reaction. Therefore, there must be the same number of atoms in the reactants as in the products of a chemical reaction. This law of science is known as the conservation of mass. It is shown by the use of balanced chemical equations.

reactant (n) a substa[nce] that reacts with anot[her] substance

product (n) a substa[nce] that is made as a res[ult] of a reaction

1 **State** the number of each type of atom in the reactants and products of the following equations. **Identify** if the equations are balanced. The first one has been done for you.

(a) C_3H_8 + $5O_2$ → $3CO_2$ + $4H_2O$

propane, oxygen, carbon dioxide, water

Reactants:

Carbon (C) = 3

Hydrogen (H) = 8

Oxygen (O) = $5 \times 2 = 10$

Products:

Carbon (C) = 3

Hydrogen (H) = $4 \times 2 = 8$

Oxygen (O) = $(3 \times 2) + 4 = 10$

Balanced/Unbalanced: balanced

(b) H_2SO_4 + $2NaOH$ → Na_2SO_4 + H_2O

sulfuric acid, sodium hydroxide, sodium sulfate, water

Reactants:

Hydrogen (H) = ____________

Sulfur (S) = ____________

Oxygen (O) = ____________

Sodium (Na) = ____________

Products:

Hydrogen (H) = ____________

Sulfur (S) = ____________

Oxygen (O) = ____________

Sodium (Na) = ____________

Balanced/Unbalanced: ____________

2 **State** whether the following chemical equations are balanced or unbalanced.

(a)

SO_3	+	H_2O	→	H_2SO_4
sulfur trioxide		water		sulfuric acid

Balanced/Unbalanced: ____________________

(b)

HCl	+	Na	→	$NaCl$	+	H_2
hydrochloric acid		sodium		sodium chloride		hydrogen

Balanced/Unbalanced: ____________________

(c)

C_4H_{10}	+	$6O_2$	→	$4CO_2$	+	$5H_2O$
butane		oxygen		carbon dioxide		water

Balanced/Unbalanced: ____________________

3 Balance the following chemical equations by filling in the blanks. The first one has been done for you.

(a)

$3H_2$	+	N_2	→	2 NH_3
hydrogen		nitrogen		ammonia

(b)

______ Mg	+	P_4	→	$2Mg_3P_2$
magnesium		phosphorus		magnesium phosphide

(c)

______ HCl	+	$CuSO_4$	→	H_2SO_4	+	$CuCl_2$
hydrochloric acid		copper sulfate		sulfuric acid		copper(II) chloride

(d)

$2Al$	+	______ CuO	→	Al_2O_3	+	______ Cu
aluminium		copper(II) oxide		aluminium oxide		copper

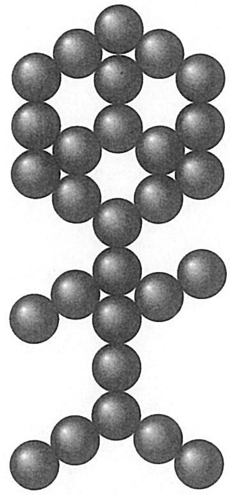
(e)

$3C$	+	______ HNO_3	→	______ CO_2	+	$2H_2O$	+	______ NO
carbon		nitric acid		carbon dioxide		water		nitric oxide

Science as a human endeavour

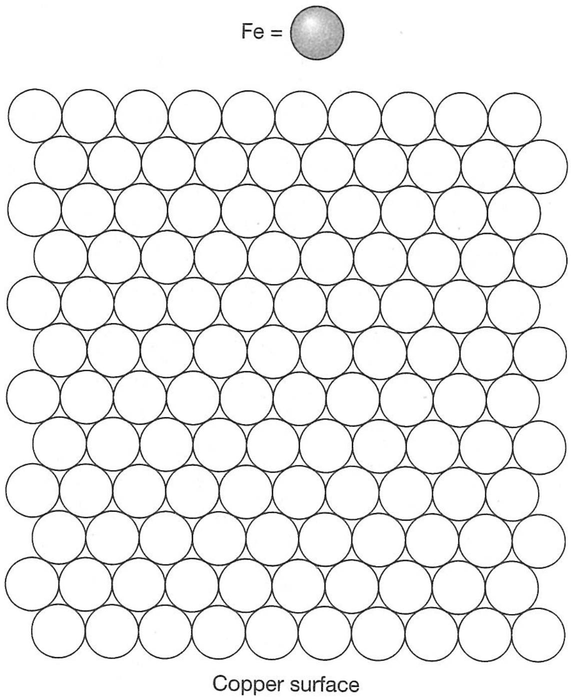
Visual/Spatial **Logical/Mathematical**

Refer to the Science as a Human Endeavour on page 146 of your student book. Today, scientists can use scanning tunnelling microscopes not only to see atoms but also to move atoms around. To demonstrate this technique, scientist have made some of the smallest pictures ever. The stick figure shown in Figure 5.2.1 was made by IBM scientists and is actually made up of carbon monoxide molecules.



Carbon Monoxide Man was made from 28 carbon monoxide molecules arranged by a scanning tunnelling microscope tip on a platinum surface. **Figure 5.2.1**

1 **Construct** your own atomic scale picture by shading in the position of your iron atoms on the copper surface represented below.



2 The diameter of an iron atom is 0.074 nm (1 nm is one-billionth of a metre). **Calculate** the height and width of your atomic scale picture. Add these measurements on your drawing.

___

___

___

3 An average human hair has a diameter of approximately 50 micrometres (50 000 nm). **Calculate** how many times your picture could fit across the width of a human hair.

___

___

___

5.3 Combustion of magnesium

Science inquiry

Verbal/Linguistic **Visual/Spatial** **Logical/Mathematical**

Before chemists understood the atomic theory of matter, people thought that the process of combustion involved the release of a material they called 'phlogiston'. Today, scientists know that phlogiston does not exist. Instead, combustion occurs when chemicals react vigorously with oxygen.

vigorously (adv) with a lot of energy

failing (n) fault; problem

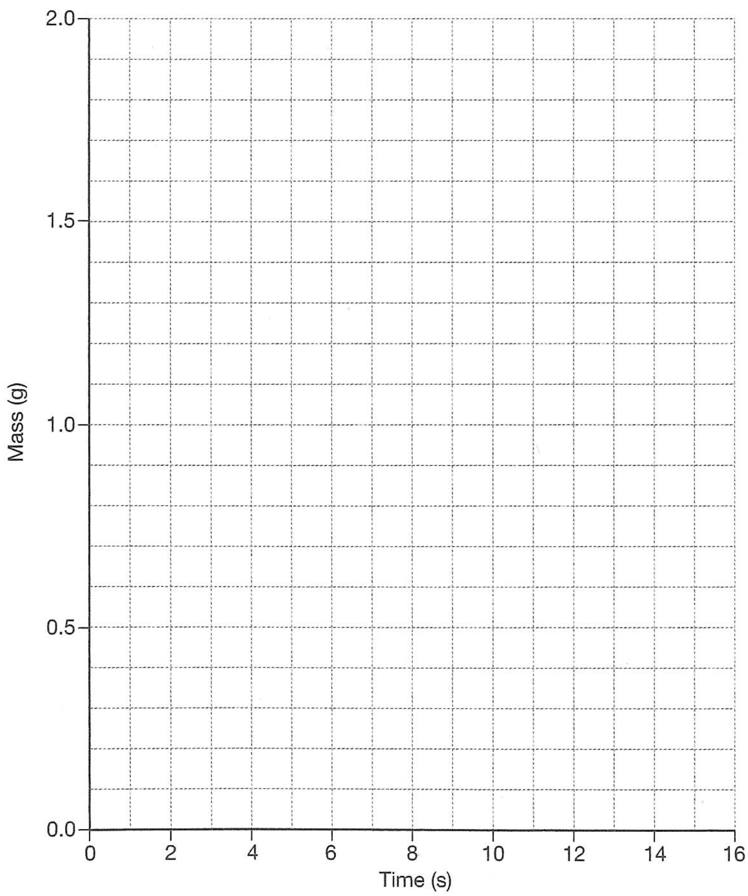
One of the failings of the phlogiston theory was that it could not explain the combustion of metals. The phlogiston theory predicted that everything that combusts should decrease in mass, due to the release of phlogiston. However, when metals burn they increase in mass. This is because they combine with oxygen to form a metal oxide, which is heavier than the metal.

Zac designed an experiment to measure the mass of 1 g of magnesium (Mg) as it burnt in oxygen (O_2) to produce magnesium oxide (MgO). His results are summarised in the tables below.

Time (s)	0	1	2	3	4	5	6	7	8
Mass (g)	1.00	1.04	1.08	1.12	1.16	1.20	1.24	1.29	1.34

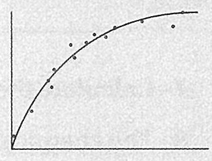
Time (s)	9	10	11	12	13	14	15	16
Mass (g)	1.39	1.44	1.49	1.55	1.61	1.67	1.67	1.67

1 Plot the data points on the axes below to **construct** a graph. Draw a line of best fit.



HINT

A 'line of best fit' is a smooth line that goes near most of the points. For example:



2 **Construct** particle model diagrams showing the atoms in the reactants and products before and after the reaction in Zac's experiment.

Before	After
Mg	
O_2	MgO

3 **Construct** a word equation and formula equation for the chemical reaction.

Word equation: magnesium + ______ $\longrightarrow$ ______

Formula equation: ______

4 **Describe** how the mass changed as the magnesium burnt in oxygen.

______

______

5 (a) From the graph, **state** how much time the chemical reaction took to complete.

The reaction took ______ seconds.

(b) **Explain** your answer.

______

______

6 (a) **State** the law of conservation of mass.

The law of conservation of mass states that atoms cannot be ______ or ______ .

(b) **Explain** how it applies to this experiment.

In this experiment ______

______

7 **Calculate** the mass of oxygen that was used in the reaction: ______ g

8 The chemical formula for magnesium oxide, MgO, indicates that there are an equal number of magnesium and oxygen atoms in the compound. **Calculate** how much heavier a magnesium atom is compared to an oxygen atom using the equation:

$$\frac{\text{Mass of Mg in MgO}}{\text{Mass of O in MgO}} = \_\_\_\_\_\_ = \_\_\_\_\_\_$$

9 Given that the atomic mass of an oxygen atom is 16, **calculate** the atomic mass of a magnesium atom in the space below.

HINT

Use your answer from question 8.

5.4 Classifying chemical reactions

Science understanding

Logical/Mathematical

Scientists classify chemical reactions into different types. The classification might depend on the type of reactants that are involved, the type of products that are produced or how the reactants interact to form the products. This helps scientists to understand more about chemical reactions because by understanding one chemical reaction, they can also understand other chemical reactions of the same type.

interact (v) to act on each other

aqueous (adj) liquid

donate (v) to give

Some common types of chemical reactions are:

- *decomposition*—when one reactant breaks apart to form two or more products
- *combination*—when two or more reactants come together to form one product
- *precipitation*—when two aqueous solutions mix to produce a solid
- *metal displacement*—when a more reactive metal donates its electrons to the ions of a less reactive metal in solution. The less reactive metal ion is reduced to form the metal atom. The reactive metal is oxidised to form ions. This is called a redox reaction.

Use these definitions to **classify** each reaction by placing a cross in the relevant column. Remember, a chemical reaction may fit more than one classification.

		Decomposition	Combination	Precipitation	Metal displacement
(a)	$4HNO_3(g) \rightarrow 2H_2O(l) + 4NO_2(g) + O_2(g)$	X			
(b)	$HCl(aq) + KOH(aq) \rightarrow H_2O(l) + KCl(aq)$				
(c)	$2AgNO_3(aq) + Na_2S(aq) \rightarrow Ag_2S(s) + 2NaNO_3(aq)$				
(d)	$S(s) + O_2(g) \rightarrow SO_2(g)$				
(e)	$Zn(s) + Cu^{2+}(aq) \rightarrow Cu(s) + Zn^{2+}(aq)$				
(f)	$2H_2(g) + O_2(g) \rightarrow 2H_2O(g)$				
(g)	$SO_3(g) + H_2O(l) \rightarrow H_2SO_4(g)$				
(h)	$2KI(aq) + Pb(NO_3)_2(aq) \rightarrow 2KNO_3(aq) + PbI(s)$				
(i)	$H_2S_2O_7(l) + H_2O(l) \rightarrow 2H_2SO_4(l)$				
(j)	$Mg(s) + Sn^{2+}(aq) \rightarrow Mg^{2+}(aq) + Sn(s)$				
(k)	$BaCl_2(aq) + Na_2SO_4(aq) \rightarrow BaSO_4(s) + 2NaCl(aq)$				
(l)	$H_2O_2(l) \rightarrow H_2(g) + O_2(g)$				
(m)	$Ca(s) + Ni^{2+}(aq) \rightarrow Ca^{2+}(aq) + Ni(s)$				

Extracting gold

Science understanding

Verbal/Linguistic

Australia is one of the world's largest exporters of gold. Before gold can be used, it first needs to be extracted. Extraction of gold from the ore (rock) occurs in the following stages.

extraction (n) remov taking out

ore (n) rock that contains minerals

1. Crushing

Rock must first be broken into smaller pieces so that it is easier to transport and treat. Large rocks are smashed into smaller pieces by a large swinging weight. The crushed rocks are now about 10 cm in diameter. The crushed rock is then fed onto a flat rubber belt called a conveyor, which transports it away for the next stage of treatment.

2. Grinding

The small rocks are then broken down to a much finer size, smaller than sand grains. This process is called grinding. Grinding is done in large rotating cylinders called ball mills. The ball mills contain many steel balls inside that roll around and grind the rock into a fine powder. Water is added to flush the fine particles out of the mill. The mixture, which is called a slurry, is pumped to the next stage.

rotating (adj) turning around

flush (v) to clean out with water

3. Froth flotation

The slurry is added to tanks called froth flotation cells. In these cells, the slurry is mixed with special chemicals called surfactants, and with air. Air is blown in at the bottom of the cell and rises up through the slurry. The surfactants stick to the fine gold and rock particles and to the rising air bubbles. Therefore, the fine particles of gold and rock float in a froth to the top of the cell. The froth is collected from the top of the cell and is then heated to a high temperature in a process called roasting. This removes sulfur impurities from the concentrated ore.

froth (n) bubbles; foa

impurity (n) somethir that makes a substar not pure

4. Carbon in leach (CIL)

After roasting, the concentrated ore is added to special tanks where it is mixed with a very poisonous chemical called sodium cyanide. The cyanide dissolves the gold out of the rock particles. The cyanide solution is then passed over tanks containing columns of carbon, which have a very large surface area. The gold solution sticks to the carbon surface.

5. Electrowinning and bullion formation

Caustic soda and sometimes cyanide are used to wash the gold solution off the carbon. This process is called elution.

The last stage of extraction is called electrowinning. This process uses electrolysis to remove gold from the gold solution. The gold collects as solid metal on steel plates. Once all the gold has been deposited on the plates, the plates are removed from the electrolytic cells. The solid gold is removed from the plates, washed, and then dried in large ovens to form a material called 'cake'.

LB 5.5 p.139

The cake is heated in a furnace, which melts the gold. The molten gold is later poured into moulds to make gold bars called bullion, which can be 65–80% gold. This bullion may later be refined to an even higher purity.

furnace (n) a very hot fire

molten (adj) melted

LB 5.5 p.139

1 **List** the five main steps in gold extraction from the mined ore.

2 **Explain** the need for grinding and describe the process. (part 2)

HINT

Why is it necessary?
How is it done?

3 **Describe** the flotation process. (part 3)

4 **Describe** the CIL process. (part 4)

5 **Identify** the two main steps in the gold recovery stage. (part 5)

6 **Explain** how gold is removed from the elution solution. (part 5)

Reaction rate and surface area of reactants

Science inquiry

Verbal/Linguistic **Visual/Spatial** **Logical/Mathematical**

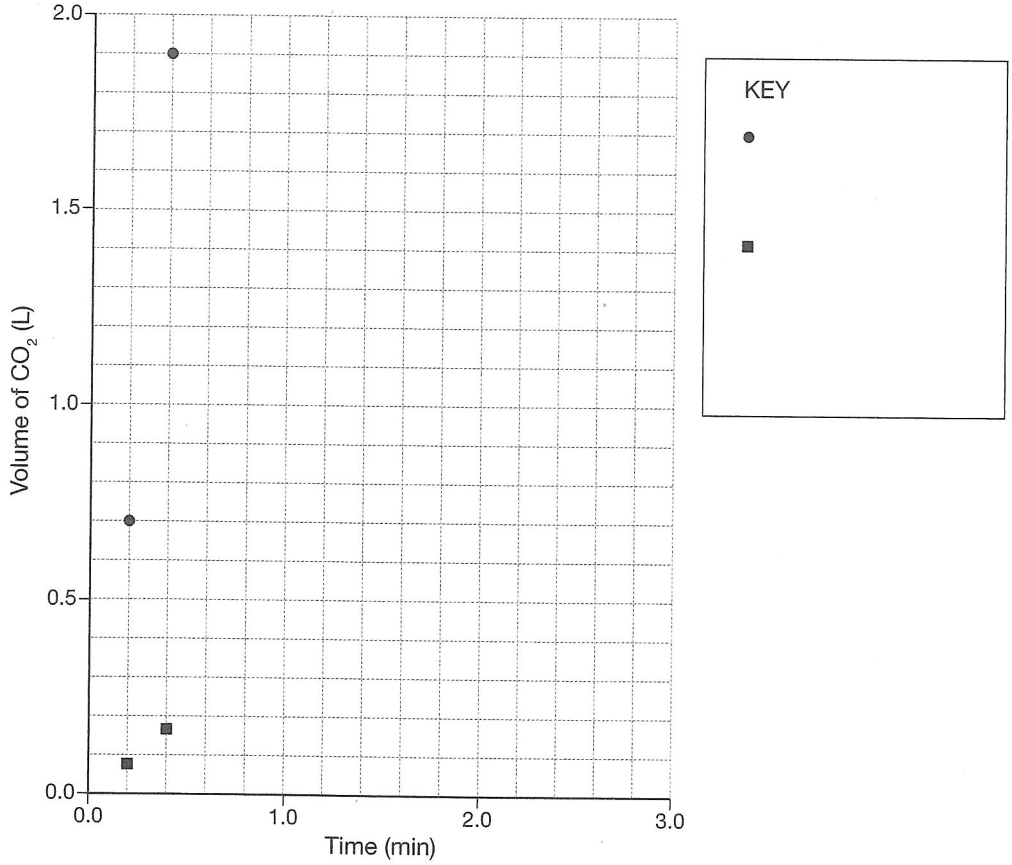
Katrina sets up an experiment to compare how crushing a soluble aspirin tablet affects the rate at which it dissolves. She places a whole aspirin tablet and a crushed aspirin tablet in separate bottles of water and measures the volume of carbon dioxide that is produced over 3 minutes. Her results are summarised in the tables below.

soluble (adj) dissolve in water

Time (min)	0.0	0.2	0.4	0.6	0.8	1.0	1.2	1.4
Volume of CO_2 Control (L)	0.00	0.08	0.16	0.26	0.36	0.46	0.58	0.70
Volume of CO_2 Crushed (L)	0.00	0.70	1.90	1.90	1.90	1.90	1.90	1.90

Time (min)	1.6	1.8	2.0	2.2	2.4	2.6	2.8	3.0
Volume of CO_2 Control (L)	0.84	0.98	1.14	1.31	1.49	1.69	1.90	1.90
Volume of CO_2 Crushed (L)	1.90	1.90	1.90	1.90	1.90	1.90	1.90	1.90

1 **Use** different symbols to **construct** two graphs from the data in Katrina's experiment on the axes below. Complete the key to show which symbol matches which data. The first two stages have been done for you.



LB 5.6 p.140

2 **Compare** how the whole aspirin tablet and the crushed aspirin tablet reacted when placed in water.

3 **Identify** whether the crushed tablet produced more, less or the same amount of carbon dioxide as the whole tablet.

4 **State** how much carbon dioxide gas was produced by each tablet.

5 From the graph, **calculate** the time it took:

(a) for the whole tablet to react completely

(b) for the crushed tablet to react completely

6 How much faster did the crushed tablet react compared to the whole tablet? **Calculate** the answer below.

7 A crushed tablet would help to get the aspirin into the bloodstream quickly and relieve pain faster. **Propose** a reason why this soluble aspirin tablet is not taken as a powder or in capsules.

HINT

Powders and capsules dissolve very quickly in the stomach. What would happen in your stomach if the aspirin dissolved very quickly?

8 Katrina learns that carbon dioxide is produced when sodium bicarbonate ($NaHCO_3$) and citric acid ($C_6H_8O_7$) in the aspirin tablet react to form sodium citrate ($NaC_6H_7O_7$), carbon dioxide (CO_2) and water (H_2O). Write the balanced formula equation for this reaction.

Science understanding

Visual/Spatial

Some chemical reactions are very fast while others are very slow. The speed of a chemical reaction is known as the rate of reaction.

Identify whether the rate of the chemical reaction is usually increased or decreased in each situation below. **Describe** what is done to control the rate.

(a) Baking a cake

Increased rate ☑ Decreased rate ☐

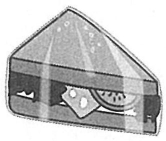
By: increasing the temperature

(b) Milk going sour

Increased rate ☐ Decreased rate ☐

By: ______

(c) Food going stale



Increased rate ☐ Decreased rate ☐

By: ______

(d) Capsules to relieve a headache

Increased rate ☐ Decreased rate ☐

By: ______

(e) Fruit ripening on the way to market

Increased rate ☐ Decreased rate ☐

By: ______

(f) Respiration during a race

Increased rate ☐ Decreased rate ☐

By: ______

(g) Starting a camp fire

Increased rate ☐ Decreased rate ☐

By: ______

(h) Converting carbon monoxide from a car exhaust into carbon dioxide

Increased rate ☐ Decreased rate ☐

By: ______

HINT

For part (h), refer to page 165 of your student book.

Literacy review

Science understanding

Verbal/Linguistic

Recall your knowledge of chemical reactions by completing the paragraphs below using words from the box.

agitation	area	balanced	catalysts	✓chemical reactions
combination	concentration	conservation	created	decomposition
destroyed	electrons	enzymes	explosion	insoluble	oxygen
precipitation	products	rate	reactants	rearranged	redox
rusting	several	single	solutions	temperature

(a) Substances can be converted into different substances through chemical reactions. The initial substances are known as the r________ and the substances that are produced are known as the p________. During a chemical reaction, the atoms in the reactants can be r________ to form the products. However, the atoms can never be c________ or d________. This is known as the law of c________ of mass. Scientists communicate this by writing b________ chemical equations.

(b) Scientists classify chemical reactions into different types. Four types of chemical reactions are d________, c________, p________ and r________ reactions. During a decomposition reaction, a s________ reactant breaks into several products. During a combination reaction, s________ reactants combine to form one product. A precipitation reaction occurs when two clear s________ mix to produce an i________ solid. In a redox reaction, o________ atoms or e________ are transferred from one substance to another.

(c) The r________ of a chemical reaction is how fast the reaction takes place. An example of a fast chemical reaction is an e________. An example of a slow chemical reaction is r________. Nature produces its own chemical helpers, called e________, to increase the rate of chemical reactions in your body.

(d) The rate of chemical reactions can be controlled by:

- controlling the t________
- controlling the c________ of the reactants
- controlling the surface a________ of the reactants
- stirring or mixing, which is also known as a________
- adding chemical helpers known as c________.